

16.4)

The force is attractive

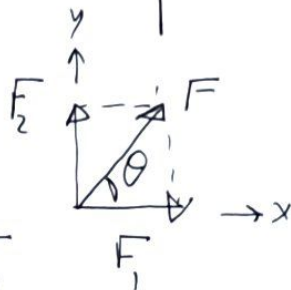
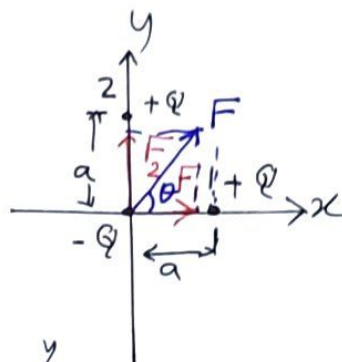
$$F_1 = k \frac{QQ}{a^2} = k \frac{Q^2}{a^2} \text{ (+ve x-axis)}$$

$$F_2 = k \frac{Q^2}{a^2} \text{ (+ve y-axis)}$$

NOTE: $F_1 = F_2$
 $\vec{F} = \vec{F}_1 + \vec{F}_2$

$$F = \sqrt{F_1^2 + F_2^2} = \sqrt{2F_1^2} = \sqrt{2} F_1$$

$$F = \sqrt{2} k \frac{Q^2}{a^2}, \text{ Direction } \tan \theta = \frac{F_2}{F_1} = 1 \Rightarrow \theta = 45^\circ$$

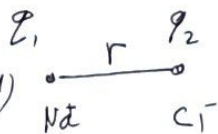


16.5)

$$q_1 = +1.6 \times 10^{-19} \text{ C}, q_2 = -1.6 \times 10^{-19} \text{ C}$$

$$F = k \frac{q_1 q_2}{r^2} = 9 \times 10^9 \frac{(1.6 \times 10^{-19})(1.6 \times 10^{-19})}{(2.3 \times 10^{-10})^2}$$

$$= 4.4 \times 10^{-9} \text{ N attractive force.}$$

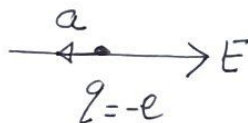


16.11)

E is opposite to a

$$F = qE = ma$$

$$\Rightarrow E = \frac{ma}{q} = \frac{9.11 \times 10^{-31} \times 10^8}{1.6 \times 10^{-19}} = 5.7 \times 10^{-17} \text{ N/C}$$

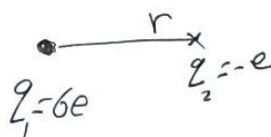


16.21)

$$V = k \frac{q}{r} = 9 \times 10^9 \frac{6 \times 1.6 \times 10^{-19}}{10^{-10}}$$

$$= 86.4 \text{ V}$$

$$U = q_2 V = -1.6 \times 10^{-19} \times 86.4 = -1.38 \times 10^{-17} \text{ J}$$

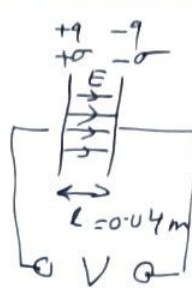


16. (23) $E = 10^4 \text{ N/C}$

16.23 $E = \frac{V}{\epsilon_0} \Rightarrow \sigma = \epsilon_0 E$

$\sigma = 8.84 \times 10^{-12} \times 10^4 = 8.84 \times 10^{-8} \text{ C/m}^2$

$V = El = 10^4 \times 0.04 = 400 \text{ V}$

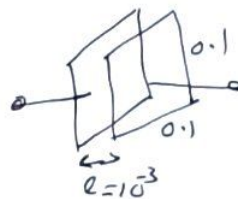


Set #1
Solution
P2

Q16.42) $C = \epsilon_0 \frac{A}{d}$

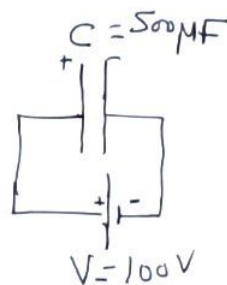
$= 8.85 \times 10^{-12} \frac{(0.1)(0.1)}{10^{-3}}$

$= 8.85 \times 10^{-11} \text{ F} = 88.5 \text{ pF}$



Q16.54) $W = U = \frac{1}{2} C V^2$

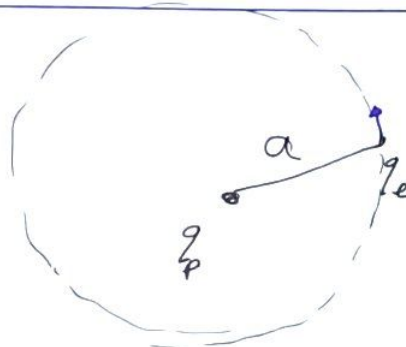
16.54 $W = \frac{1}{2} \times 500 \times 10^{-6} (100)^2 = 2.5 \text{ J}$



Q(1) $q_p = +1.6 \times 10^{-19} \text{ C}$, $q_e = -1.6 \times 10^{-19} \text{ C}$

$a = 5.3 \times 10^{-11} \text{ m} = \text{Bohr radius}$

$F = K \frac{q_p q_e}{a^2} = 9 \times 10^9 \frac{(1.6 \times 10^{-19})(1.6 \times 10^{-19})}{(5.3 \times 10^{-11})^2}$
 $= 8.2 \times 10^{-8} \text{ N}$



Q(2) $F = qE = ma$

$a = \frac{qE}{m} = \frac{8 \times 10^{-6} \times 1500}{0.005}$

$= 2.4 \text{ m/s}^2$ in the direction of $\vec{E}$

$\vec{E} = 1500 \text{ N/C}$
 $+q = 8 \mu\text{C}$
 $m = 0.005 \text{ kg}$

Q(3): $\vec{E}$ has
Constant magnitude
and constant
direction:

Q(4) $U = K \frac{q_1 q_2}{r}$
 $U = 9 \times 10^9 \frac{(+2 \times 10^{-6})(-5 \times 10^{-6})}{3}$
 $= -0.03 \text{ J}$

$q_1 = 2 \mu\text{C}$, $q_2 = -5 \mu\text{C}$